

ARUMUGAM N

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Portfolio: arumugam.qzz.io | **Available for Immediate Joining**

SUMMARY

Associate Software Engineer at TrizenAI building multi-tenant photography studio platform. Built REST APIs with NestJS/TypeScript, integrated WhatsApp Business API, and shipped AI face recognition using AWS Rekognition for automated photo delivery. Work with PostgreSQL, React, and AWS daily. Previously built ML systems including a plant identification rover (92% accuracy). Experienced in full-stack development and AI/ML integration.

EDUCATION

Francis Xavier Engineering College

Bachelor of Technology in Computer Science and Business Systems

CGPA: 8.6/10.0

Tirunelveli, Tamil Nadu

Sept 2021 – May 2025

Tilak Vidhyalaya Higher Secondary School

Higher Secondary Certificate

Percentage: 91.5%

Tirunelveli, Tamil Nadu

June 2019 – May 2021

EXPERIENCE

Associate Software Engineer

TrizenAI

April 2025 – Present

Chennai, Tamil Nadu

- Built REST APIs for Snapflo, a multi-tenant photography studio platform, working with 5+ engineers using NestJS, TypeScript, and PostgreSQL with Drizzle ORM for database schema and migrations
- Built photo sharing system using AWS Rekognition for face recognition — guests get their photos automatically via WhatsApp after face matching (S3, Server-Sent Events)
- Integrated WhatsApp Business API to automate booking confirmations, payment reminders, and cancellations using Meta compatible templates to avoid manual messaging work by 70%
- Built quotation and billing module from scratch with full status lifecycle workflow including database schema, NestJS backend, and React frontend
- Implemented RBAC, auth flows, and shipped features through CI/CD pipelines in Agile sprints with code reviews

Python Full Stack Developer Trainee

Qspiders Campus Connect

January 2025 – March 2025

Chennai, Tamil Nadu

- Built 5 full-stack applications in 8-week bootcamp using Django and Flask with MySQL, implementing MVC patterns and RESTful API design
- Built RESTful web services with JSON and JWT authentication, optimized SQL queries to make APIs 30% faster
- Used database connectivity and ORM principles, followed SDLC practices with Git version control on Linux
- Worked in Agile methodology with sprint planning, daily standups, and code reviews, delivered features on time

Software Development Intern

Quantanics Private Limited

January 2024 – March 2024

Madurai, Tamil Nadu

- Built ML web application with Flask backend, implemented RESTful APIs for ML model deployment
- Worked in Agile development with requirements analysis, system design, and testing
- Used Git for version control and code reviews, maintained code quality with unit testing
- Got hands-on with Linux systems, database connectivity, and API integration

TECHNICAL SKILLS

Backend Development: NestJS, Django, Flask, RESTful APIs, PostgreSQL, MySQL, MongoDB, Drizzle ORM

Frontend: React, TypeScript, JavaScript, HTML5, CSS3

Programming: Python, TypeScript, JavaScript, SQL, C, C++, Rust (Learning)

Machine Learning & AI: AWS Rekognition, TensorFlow, PyTorch, OpenCV, YOLOv8, Scikit-learn, NLP, Computer Vision

Cloud & DevOps: AWS (S3, EC2, Rekognition), Docker, Git, CI/CD pipelines

Core Competencies: Data Structures & Algorithms, Database Design, System Design, OOP, API Integration

Methodologies: Agile/Scrum, SDLC, Test-Driven Development, Code Reviews

PROJECTS

ZenxBill: Retail Billing Software | *Rust, SQLite, CLI, System Design*

November 2025 – Present

- Building retail billing system in Rust for photography studios with modular architecture
- Built inventory management, invoice generation, and purchase workflows with SQLite storage and CLI
- Used Rust's ownership and memory safety to get zero crashes in production
- Rewrote old Python billing code in Rust for better performance and easier maintenance

podCaster: AI-Powered Podcast Generator | *Django, Python, HTML/CSS/JS* | [GitHub](#)

June 2025 – July 2025

- Built web app that generates podcast scripts using DeepSeek R1 AI model
- Made two-step UI with topic input and chatbot workflow for creating podcasts faster
- Used prompt engineering to get natural-sounding scripts from user topics
- Added Piper TTS engine to convert markdown scripts to audio files

MedBot: AI-Integrated Rover | *Python, TensorFlow, NVIDIA Jetson, OpenCV*

January 2023 – March 2023

- Built autonomous robot using NVIDIA Jetson Nano for identifying medicinal plants — 80% faster than manual identification
- Trained deep learning model achieving 92% accuracy across 50+ plant species using custom CNN
- Optimized computer vision pipeline to process 30 FPS video for outdoor field use

Org-Assist: Organizational Chatbot | *Python, Flask, MongoDB, NLP* | [GitHub](#)

October 2023 – November 2023

- Built chatbot for organization info and complaints using Flask and MongoDB
- Used Python NLP libraries for understanding questions and generating responses
- Set up MongoDB with Flask for handling multiple users at once

ACHIEVEMENTS & CERTIFICATIONS

First Prize – Srishti International Project Exhibition for MedBot innovation (March 2023)

First Prize – Internal Hackathon, Software Domain, Francis Xavier Engineering College (October 2023)

Oracle Certified Professional – Generative AI Certification (August 2025)

NPTEL Certification – Natural Language Processing with Python Implementation (October 2023)